

SingleRAN

eCPRI Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 03 (2025-08-30).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added support for eCPRI by the UMPThb. For details, see 3.3.3 Hardware .	None	5900 series base stations (macro base stations)
Added support for eCPRI in NB-IoT. For details, see 2.2 Applicable RAT , 2.3 Differences , 3.1.2 Line Rate Requirements in Typical Cell Configurations , 3.1.3 Bandwidth-based Admission Control for Cell Setup , and 3.4.1.1 Data Preparation .	None	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	Base Station Model
Added the required line rates over an eCPRI interface, peak bandwidths, and guaranteed bandwidths of high-frequency cells that use the 2:3 slot configuration. For details, see 3.1.2 Line Rate Requirements in Typical Cell Configurations and 3.1.3 Bandwidth-based Admission Control for Cell Setup .	None	3900 and 5900 series base stations (macro base stations)

Editorial Changes

Revised the descriptions of parameter settings related to cell activation. For details, see [3.4.1.1 Data Preparation](#) and [3.4.1.2 Using MML Commands](#).

2 About This Document

2.1 General Statements

Purpose

This document is intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to UMTS, LTE FDD, LTE TDD, NB-IoT, and NR.

2.3 Differences

Table 2-1 Differences between LTE FDD and LTE TDD

Function Name	Difference	Chapter/Section
eCPRI	None	3 eCPRI

Table 2-2 Differences between NB-IoT and LTE FDD

Function Name	Difference	Chapter/Section
eCPRI	None	3 eCPRI

Table 2-3 Differences between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
eCPRI	None	3 eCPRI

Table 2-4 Differences between NSA and SA

Function Name	Difference	Chapter/Section
eCPRI	None	3 eCPRI

Table 2-5 Differences between high frequency bands and low frequency bands

Function Name	Difference	Chapter/Section
eCPRI	None	3 eCPRI

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

3 eCPRI

3.1 Principles

3.1.1 Overview

The Enhanced Common Public Radio Interface (eCPRI) protocol is used for the connection between a baseband unit (BBU) and an active antenna unit (AAU) in a radio base station. After a BBU is connected to an AAU through their CPRI ports, a link between them is then set up to transmit control-plane, user-plane, and clock data. In the following sections, an eCPRI interface or port refers to a CPRI interface or port using the eCPRI protocol.

The eCPRI protocol differs from the CPRI protocol in two aspects:

- eCPRI enables functional split at the physical layer. Specifically, certain processing units at the physical layer are relocated from the BBU to the AAU. For details, see sections 2.3 "Functional Decomposition" and 6.1.1 "eCPRI Functional Decomposition" in eCPRI Specification V1.1.
- eCPRI uses the Ethernet for data transmission, whereas CPRI uses the time division multiplexing (TDM) for data transmission. For details, see section 3.2.1 "User Plane over Ethernet" in eCPRI Specification V1.1.

For details about the networking modes supported by the eCPRI protocol, see *RF Unit and Topology Management*.

3.1.2 Line Rate Requirements in Typical Cell Configurations

Requirements for the line rate over an eCPRI interface are related to the cell bandwidth, number of antennas, numbers of uplink and downlink layers, and other factors. [Table 3-1](#), [Table 3-2](#), [Table 3-3](#), [Table 3-4](#), [Table 3-5](#), [Table 3-6](#), and [Table 3-7](#) list requirements for the line rate over an eCPRI interface under typical cell configurations.

Table 3-1 Requirements for the line rate over an eCPRI interface under typical NR TDD (low-frequency) cell configurations (16 layers in the downlink; 8 layers in the uplink)

Bandwidth	Line Rate (64T64R)	Line Rate (32T32R)
100 MHz	25 Gbit/s	25 Gbit/s
90 MHz	22.5 Gbit/s	22.5 Gbit/s
80 MHz	20 Gbit/s	20 Gbit/s
70 MHz	17.5 Gbit/s	17.5 Gbit/s
60 MHz	15 Gbit/s	15 Gbit/s
50 MHz	12.5 Gbit/s	12.5 Gbit/s
40 MHz	10 Gbit/s	10 Gbit/s
30 MHz	7.5 Gbit/s	7.5 Gbit/s
20 MHz	5 Gbit/s	5 Gbit/s

Table 3-2 Requirements for the line rate over an eCPRI interface under NR TDD (high-frequency) cell configurations

Bandwidth	Line Rate (4T4R)	Line Rate (8T8R)
200 MHz; 4:1 slot configuration (4_1_DDDSU)	6.25 Gbit/s	12.5 Gbit/s
200 MHz; 2:3 slot configuration (2_3_DSUUU)	12.5Gbit/s	N/A
100 MHz; 4:1 slot configuration (4_1_DDDSU)	3.125 Gbit/s	6.25 Gbit/s
100 MHz; 2:3 slot configuration (2_3_DSUUU)	6.25Gbit/s	N/A

Table 3-3 Requirements for the line rate over an eCPRI interface under typical NR FDD cell configurations

Bandwidth	Line Rate (32T32R)
50 MHz	12.5 Gbit/s
40 MHz	10 Gbit/s

Bandwidth	Line Rate (32T32R)
30 MHz	7.5 Gbit/s
20 MHz	5 Gbit/s
15 MHz	3.75 Gbit/s
10 MHz	2.5 Gbit/s

Table 3-4 Requirements for the line rate over an eCPRI interface under typical LTE TDD cell configurations (16 layers in the downlink; 8 layers in the uplink)

Bandwidth	Line Rate (64T64R)	Line Rate (32T32R)
20 MHz	5 Gbit/s	5 Gbit/s
15 MHz	3.75 Gbit/s	3.75 Gbit/s
10 MHz	2.5 Gbit/s	2.5 Gbit/s

Table 3-5 Requirements for the line rate over an eCPRI interface under typical LTE FDD cell configurations

Bandwidth	Line Rate (32T32R)
20 MHz	5 Gbit/s
15 MHz	3.75 Gbit/s
10 MHz	2.5 Gbit/s
5 MHz	1.25 Gbit/s

Table 3-6 Requirements for the line rate over an eCPRI interface under typical NB-IoT cell configurations

Bandwidth	Line Rate (32T32R)
200KHz	0.25Gbit/s

Table 3-7 Requirements for the line rate over an eCPRI interface under typical UMTS cell configurations

Bandwidth	Line Rate (32T32R)
5 MHz	0.5 Gbit/s

 NOTE

- The one-way transmission delay over an eCPRI interface between a BBU and an AAU must be less than or equal to 100 μ s. The maximum distance between a BBU and an AAU is 20 km when no transmission equipment is used between them. The maximum distance is less than 20 km when transmission equipment is used between them. This is because extra delay is generated due to the usage of transmission equipment. If the distance exceeds 20 km, the cell capabilities, such as uplink and downlink throughput and access success rate decrease.
- In the current version, 10 Gbit/s, 25 Gbit/s, and 50 Gbit/s optical modules are supported for UMTS, LTE FDD, LTE TDD, (low-frequency) NR TDD, and NR FDD, allowing only optical fiber connections of 1x10 Gbit/s, 1x25 Gbit/s, 1x50 Gbit/s, 2x10 Gbit/s, 2x25 Gbit/s, and 2x50 Gbit/s, but not the optical fiber connection of 1x10 Gbit/s+1x25 Gbit/s, 1x10 Gbit/s+1x50 Gbit/s, and 1x25 Gbit/s+1x50 Gbit/s. 25 Gbit/s and 50 Gbit/s optical modules are supported for (high-frequency) NR TDD, allowing only optical fiber connections of 1x25 Gbit/s, 1x50 Gbit/s, 2x25 Gbit/s, and 2x50 Gbit/s. When the eCPRI interface configuration does not meet the interface bandwidth requirement, cell setup may fail, or the numbers of uplink and downlink layers or the traffic specifications in the cell may be limited.
- According to LTE TDD bandwidth requirements, a single 25 Gbit/s optical fiber can bear a maximum data volume of five 20 MHz 64T64R cells. If cell setup fails, check the number of carriers on the optical fibers first.
 1. Run the **LST RRUCCHAIN** command to view the number of optical fibers.
 2. Run the **DSP CELLPHYTOPO** command to query the carrier configuration of an AAU. Ensure that the number of carriers is less than or equal to five times the number of optical fibers.
- When an eCPRI interface serves multiple cells, the required line rate is the sum of line rates required by all cells carried on the eCPRI interface.
- For a cell, the required line rate over an eCPRI interface is directly proportional to the number of layers and cell bandwidth.

3.1.3 Bandwidth-based Admission Control for Cell Setup

When multiple cells served by an AAU share optical fibers, the optical fiber bandwidth may not meet the cell bandwidth requirements. The cells can be multiple NR cells, LTE cells, UMTS and LTE cells sharing an AAU, UMTS and NR cells sharing an AAU, UMTS, LTE, and NR cells sharing an AAU, or NR and LTE cells sharing an AAU. Provided that the peak bandwidths of multiple cells are not concurrent, admission control during cell setup is performed based on the bandwidth lower than the peak bandwidth. [Table 3-8](#) describes the admission policies for NR TDD cells. [Table 3-9](#) describes the admission policies for NR FDD cells. [Table 3-10](#) describes the admission policies for LTE TDD cells. [Table 3-11](#) describes the admission policies for LTE FDD cells. [Table 3-12](#) describes the admission policies for NB-IoT cells. [Table 3-14](#) describes the admission policies for UMTS cells.

For LTE TDD cells, the eCPRI compression function can be used to reduce the single-carrier admission bandwidth, increasing the number of carriers supported on an eCPRI optical fiber and lowering the optical fiber deployment cost. The eCPRI compression function takes effect when the **Cell.CPRICompression** parameter is set to **ECPRI_COMPRESSION**. This function applies only to massive MIMO cells. [Table 3-13](#) describes the admission policies for LTE TDD cells after the function takes effect. eCPRI compression is not supported in LTE FDD.

During cell setup, the base station applies for bandwidth based on the guaranteed bandwidth. If the bandwidth application fails, the cell setup fails. If the actual data

transmission requirements of cells are higher than the actual transmission bandwidth of optical fibers, the numbers of uplink and downlink layers or traffic specifications of cells are limited.

It is recommended that the function of reporting an alarm indicating insufficient eCPRI/CPRI resources be enabled in NR 8T8R and massive MIMO scenarios to solve the problem that no alarm was reported when eCPRI/CPRI resources are insufficient in a cell. This function is controlled by the **ECPRI_CPRI_INSUFT_ALM_RPT_SW** option of the **gNBOamParam.AlarmSwitch** parameter. If this option is selected, ALM-29871 NR DU Cell TRP Capability Degraded is reported when the eCPRI bandwidth does not meet the maximum service requirement or 1/8 or fewer channels have not been allocated due to insufficient cell CPRI bandwidth or backplane bandwidth. If this option is deselected, this alarm is not reported in the preceding scenario.

Table 3-8 Peak bandwidth and guaranteed bandwidth of NR TDD cells

Cell Type	UL/DL Peak Bandwidth (Gbit/s)	UL/DL Guaranteed Bandwidth (Gbit/s)
Low-frequency 100 MHz 64T64R/32T32R	25	5
Low-frequency 90 MHz 64T64R/32T32R	22.5	5
Low-frequency 80 MHz 64T64R/32T32R	20	5
Low-frequency 70 MHz 64T64R/32T32R	17.5	5
Low-frequency 60 MHz 64T64R/32T32R	15	5
Low-frequency 50 MHz 64T64R/32T32R	12.5	5
Low-frequency 40 MHz 64T64R/32T32R	10	5
Low-frequency 30 MHz 64T64R/32T32R	7.5	5
Low-frequency 20 MHz 64T64R/32T32R	5	5
High-frequency 200 MHz 8T8R; 4:1 slot configuration (4_1_DDDSU)	12.5	12.5
High-frequency 200 MHz 4T4R; 4:1 slot configuration (4_1_DDDSU)	6.25	6.25

Cell Type	UL/DL Peak Bandwidth (Gbit/s)	UL/DL Guaranteed Bandwidth (Gbit/s)
High-frequency 200 MHz 4T4R; 2:3 slot configuration (2_3_DSUUU)	12.5	12.5
High-frequency 100 MHz 8T8R; 4:1 slot configuration (4_1_DDDSU)	6.25	6.25
High-frequency 100 MHz 4T4R; 4:1 slot configuration (4_1_DDDSU)	3.125	3.125
High-frequency 100 MHz 4T4R; 2:3 slot configuration (2_3_DSUUU)	6.25	6.25

Table 3-9 Peak bandwidth and guaranteed bandwidth of NR FDD cells

Cell Type	UL/DL Peak Bandwidth (Gbit/s)	UL/DL Guaranteed Bandwidth (Gbit/s)
40 MHz 32T32R	10	1.25
30 MHz 32T32R	7.5	1.25
25 MHz 32T32R	6.25	1.25
20 MHz 32T32R	5	1.25
15 MHz 32T32R	3.75	1.25
10 MHz 32T32R	2.5	1.25

Table 3-10 Peak bandwidth and guaranteed bandwidth of LTE TDD cells

Cell Type	UL/DL Peak Bandwidth (Gbit/s)	UL/DL Guaranteed Bandwidth (Gbit/s)
20 MHz 64T64R/32T32R	5	5
15 MHz 64T64R/32T32R	3.75	3.75
10 MHz 64T64R/32T32R	2.5	2.5

Table 3-11 Peak bandwidth and guaranteed bandwidth of LTE FDD cells

Cell Type	UL/DL Peak Bandwidth (Gbit/s)	UL/DL Guaranteed Bandwidth (Gbit/s)
20 MHz 32T32R	5	5
15 MHz 32T32R	3.75	3.75
10 MHz 32T32R	2.5	2.5
5 MHz 32T32R	1.25	1.25

Table 3-12 Peak bandwidth and guaranteed bandwidth of NB-IoT cells

Cell Type	UL/DL Peak Bandwidth (Gbit/s)	UL/DL Guaranteed Bandwidth (Gbit/s)
200KHz 32T32R	0.25	0.25

Table 3-13 Peak bandwidth and guaranteed bandwidth of LTE TDD cells after the eCPRI compression function takes effect

Cell Type	UL/DL Peak Bandwidth (Gbit/s)	UL/DL Guaranteed Bandwidth (Gbit/s)
20 MHz 64T64R/32T32R	5	3
15 MHz 64T64R/32T32R	3.75	2.25
10 MHz 64T64R/32T32R	2.5	1.5

Table 3-14 Peak bandwidth and guaranteed bandwidth of UMTS cells

Cell Type	UL/DL Peak Bandwidth (Gbit/s)	UL/DL Guaranteed Bandwidth (Gbit/s)
5 MHz 32T32R	0.5	0.5

3.1.4 eCPRI MUX

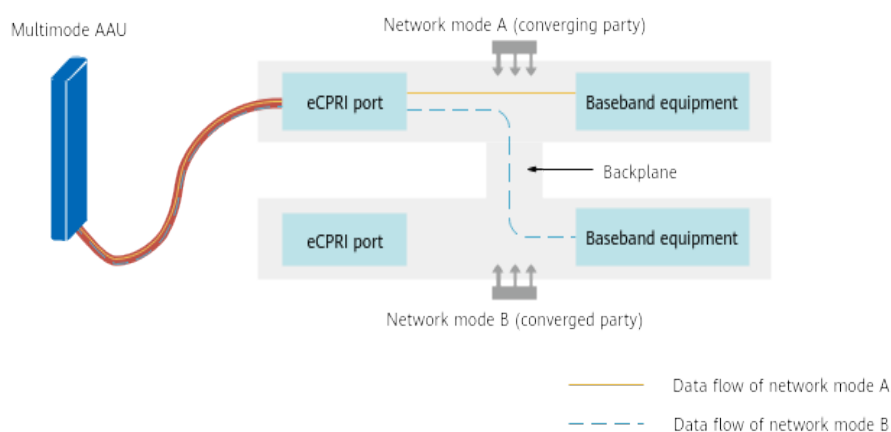
eCPRI Multiplex (eCPRI MUX) enables multiple modes in a multimode base station to use the same eCPRI port for data transmission. eCPRI MUX can only be used within one BBU.

eCPRI MUX is implemented by converging data of multiple modes. [Figure 3-1](#) shows the eCPRI MUX principles. The board of network mode B is connected to the board of network mode A through the backplane. In this way, data of network mode B is converged on the board of network mode A. Then, the board of

network mode A is connected to a multimode AAU through an optical fiber which carries data of both network modes A and B.

The board of network mode A serves as the converging board and network mode A is the converging party. The board of network mode B serves as the converged board and network mode B is the converged party.

Figure 3-1 eCPRI MUX principles when data of multiple network modes is converged

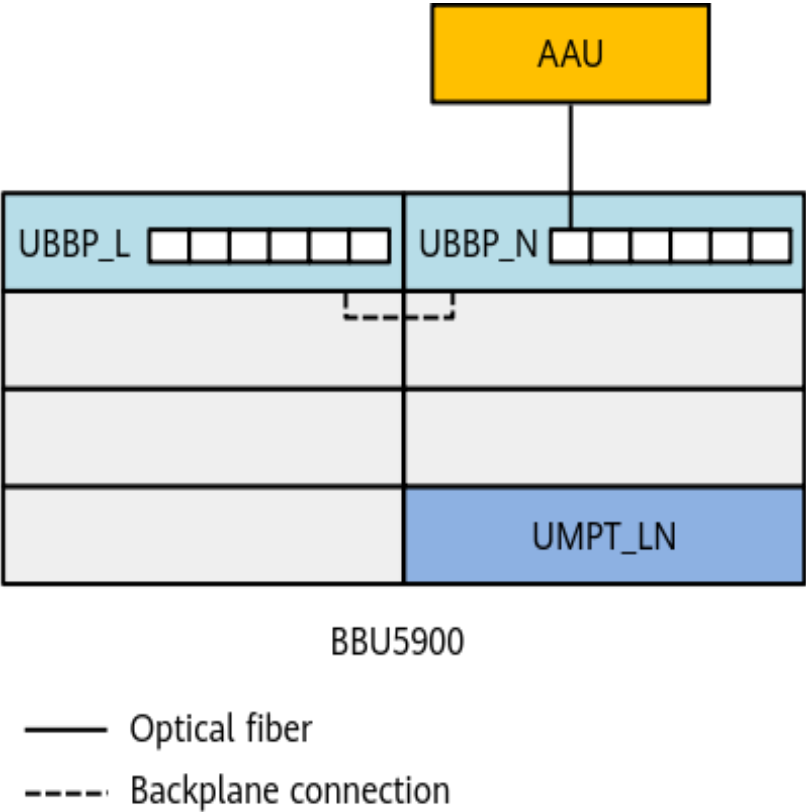


3.1.4.1 Co-MPT Scenarios

LTE and NR Co-MPT

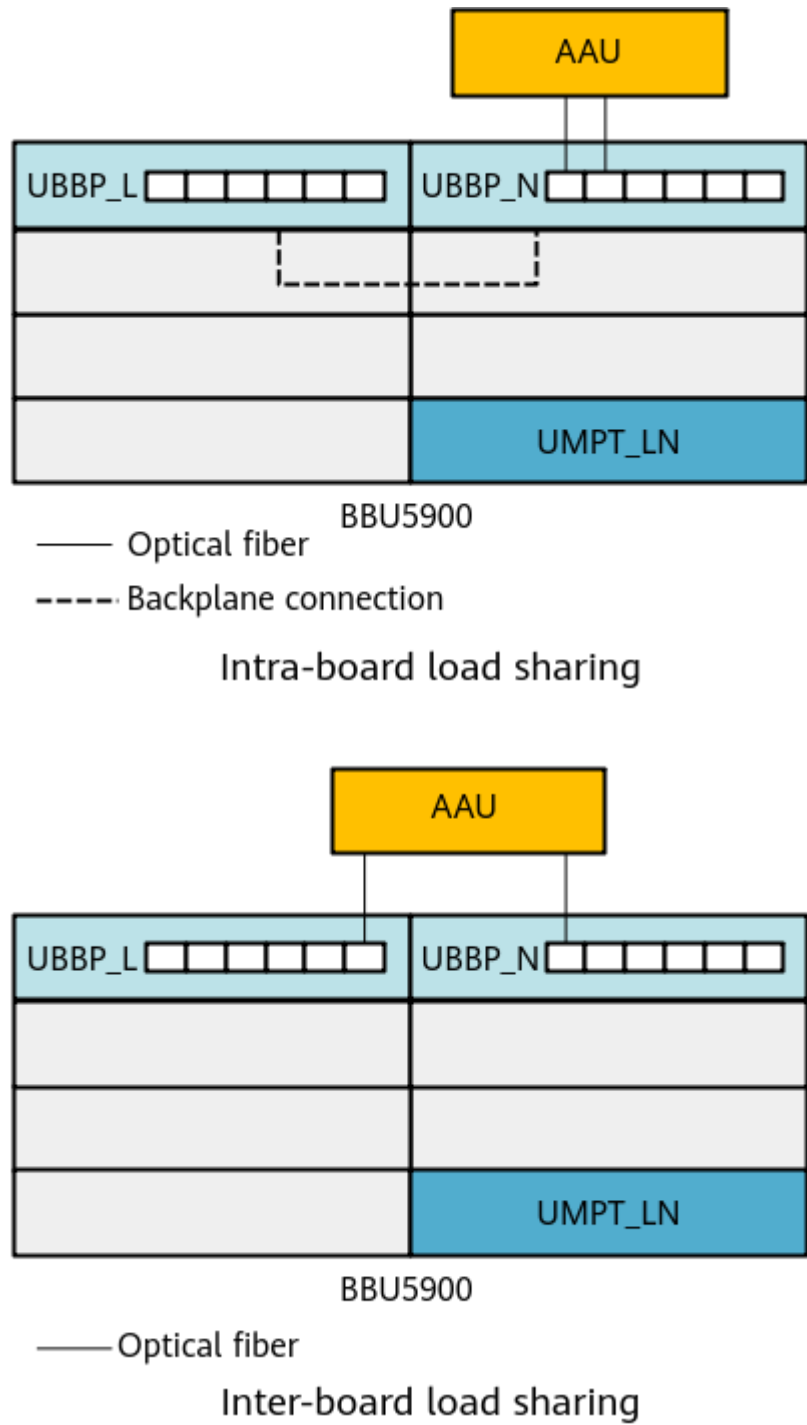
In LTE and NR co-MPT scenarios, when the total transmission bandwidth required by the LTE and NR cells served by an AAU is less than the transmission bandwidth of the optical fiber connecting to the AAU, eCPRI MUX can be used. The AAU can be connected to an LTE baseband processing unit or an NR baseband processing unit through one optical fiber. A single-fiber eCPRI link is set up between the BBU and AAU through a baseband processing unit for the transmission of control-plane, user-plane, and clock data, as shown in [Figure 3-2](#). This figure takes the use of one converged board as an example, and the use of multiple converged boards is also supported.

Figure 3-2 Single-fiber eCPRI MUX in a co-MPT scenario (using the BBU5900 as an example)



A load sharing topology is used when the physical bandwidth of an eCPRI link is insufficient to support a cell. In a load sharing topology, a BBU is connected to an AAU through two interfaces. This ensures that the physical bandwidth of the two interfaces can meet the cell bandwidth requirements. Two eCPRI links are set up between the BBU and AAU. User-plane and clock data is transmitted on both links. Control-plane data is transmitted on either of the links. Load sharing can be classified into intra-board load sharing and inter-board load sharing, as shown in [Figure 3-3](#). This figure takes the use of one converged board in an intra-board load sharing topology as an example, and the use of multiple converged boards is also supported.

Figure 3-3 eCPRI MUX with a load sharing topology in a co-MPT scenario (using the BBU5900 as an example)



- Intra-board load sharing (recommended)
 - When the bandwidth of one eCPRI link is sufficient for all services, the other eCPRI link provides redundancy backup.
 - When one eCPRI link is faulty, the services carried on the link are automatically switched over to the other eCPRI link.

- When one eCPRI link is faulty and the bandwidth of the other eCPRI link is insufficient to support all cells, some cells may become unavailable.
- Inter-board load sharing
 - When the eCPRI link that carries O&M services is faulty, the O&M services are automatically switched over to the other eCPRI link.
 - Enhanced eCPRI-based inter-board load sharing is controlled by the **EQUIPMENT.ECPRIIBENHSW (LTE eNodeB, 5G gNodeB)** parameter. With this parameter set to **ON**, if one of the eCPRI links that carry user-plane or O&M services is faulty and the admission bandwidth is sufficient, services carried on the faulty link are migrated to the other eCPRI link. After the link fault is rectified, these services are automatically migrated back to the original link. With this parameter set to **OFF**, only O&M services are automatically switched over to the other eCPRI link. Each eCPRI link carries only user-plane services of the baseband processing unit connected to it.
 - In the multimode inter-board load sharing topology, two ports on an AAU must connect to two baseband processing units in different modes. Services of each baseband processing unit directly connected to the AAU through an optical fiber are carried only on the eCPRI link connected to this baseband processing unit. Services of each baseband processing unit not directly connected to the AAU through an optical fiber are carried only on the eCPRI link that is connected to the baseband processing unit directly connected to the AAU through an optical fiber in the same mode.

NOTE

- It is not recommended that two baseband processing units in the same mode be used for inter-board load sharing. In this scenario, cells must be deployed on the baseband processing units that directly connect to the AAU through optical fibers.
- When the O&M services carried on a faulty eCPRI link are automatically switched over to the other eCPRI link, service data loss lasts for a while.

UL/UN/ULN Co-MPT

When eCPRI MUX is used in UL (UMTS and LTE FDD)/UN (UMTS and NR FDD)/ULN (UMTS, LTE FDD, and NR FDD) co-MPT scenarios, the AAU can be connected to an LTE baseband processing unit or an NR baseband processing unit through an optical fiber. A single-fiber eCPRI link is set up between the BBU and AAU through a baseband processing unit for the transmission of control-plane, user-plane, and clock data. When the **RRUCHAIN.PROTOCOL (LTE eNodeB, 5G gNodeB)** parameter is set to **eCPRI** for a UBBPg series board serving as the converging board, and the working mode specified by the **RRU.RS (LTE eNodeB, 5G gNodeB)** parameter is changed from excluding UMTS to including UMTS or vice versa, the board automatically resets to validate the new configuration. If the working mode specified by the **RRU.RS (LTE eNodeB, 5G gNodeB)** parameter includes UMTS, the eCPRI link of the RF module fixedly occupies the bandwidths of two UMTS cells.

The constraints on eCPRI MUX in this scenario are listed as follows:

- UMTS can only serve as the converged party, but cannot serve as the converging party.
- The converging party supports only FDD but does not support TDD.

- The converging boards must be UBBPg or UBBPh series boards, and the converged boards must be UBBP series boards.
- Only single-fiber eCPRI MUX is supported. The load sharing topology is not supported.
- A UBBPg series board serving as the converging board can forward data of a maximum of six UMTS cells. A UBBPh series board serving as the converging board can forward data of a maximum of 18 UMTS cells. (The actual number of cells whose data can be forwarded depends on the number of board ports compatible with DSFP optical modules.)

Figure 3-4, Figure 3-5, and Figure 3-6 show eCPRI MUX in UL, UN, and ULN co-MPT scenarios, respectively.

Figure 3-4 Single-fiber eCPRI MUX in a UL co-MPT scenario (using the BBU5900 as an example)

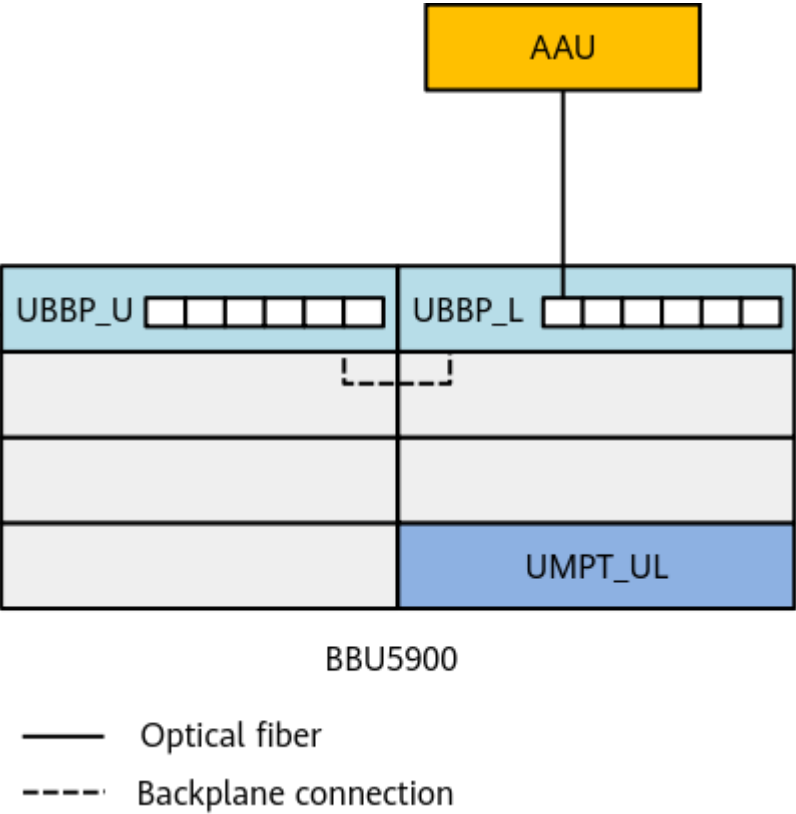


Figure 3-5 Single-fiber eCPRI MUX in a UN co-MPT scenario (using the BBU5900 as an example)

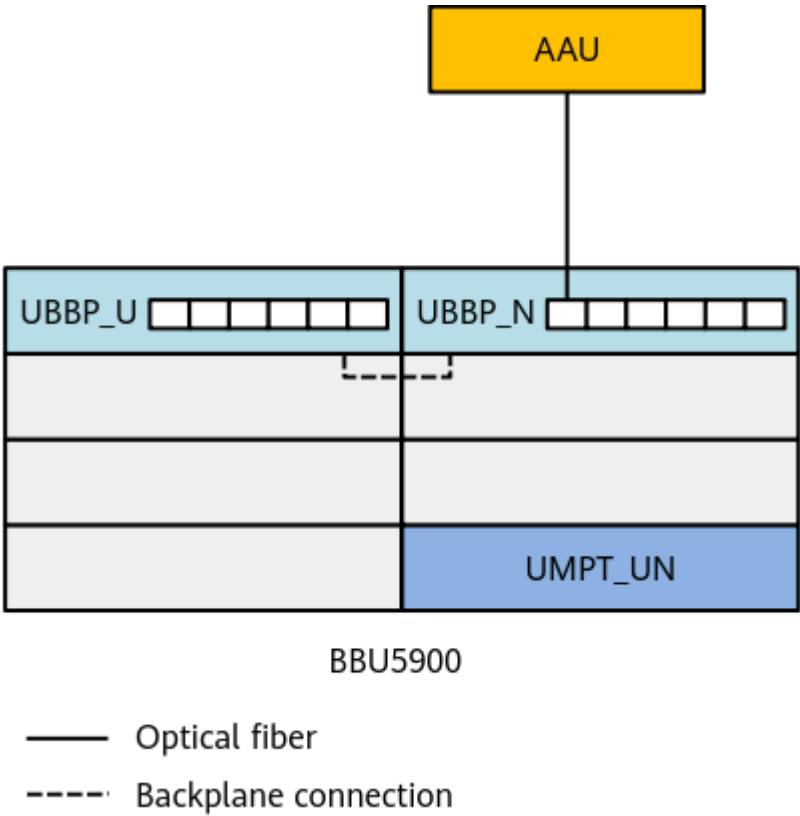
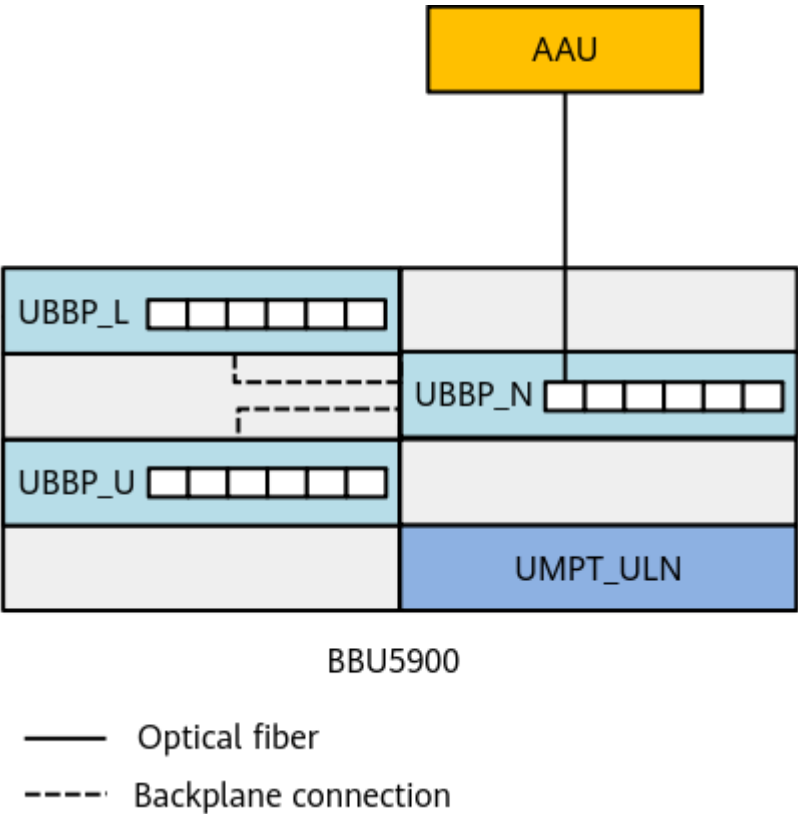


Figure 3-6 Single-fiber eCPRI MUX in a ULN co-MPT scenario (using the BBU5900 as an example)

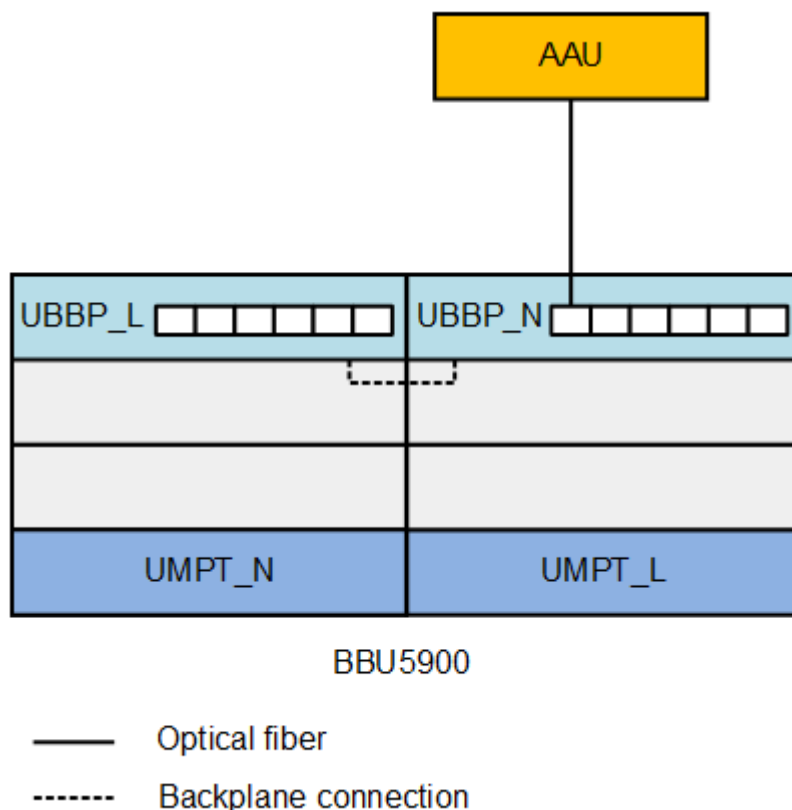


3.1.4.2 Separate-MPT Scenarios

LTE and NR Separate-MPT

In LTE and NR separate-MPT scenarios, when the total transmission bandwidth required by the LTE and NR cells served by an AAU is less than the transmission bandwidth of the optical fiber connecting to the AAU, eCPRI MUX can be used. The AAU can be connected to an LTE or NR baseband processing unit through an optical fiber, as shown in [Figure 3-7](#).

Figure 3-7 eCPRI MUX in a separate-MPT scenario (using the BBU5900 as an example)



The constraints on eCPRI MUX in separate-MPT scenarios are listed as follows:

- The converging party cannot serve as the converged party.
- The AAU and the corresponding chain/ring on the converging party must be configured prior to those on the converged party. If the AAU and the corresponding chain/ring on the converging party is not configured, services of the converged party are unavailable.
- Each BBU supports three pairs of one-to-one convergence relationships, that is, a maximum of 18 ports can be provided for data convergence. One converged board cannot correspond to two converging boards, and two converged boards cannot correspond to one converging board.
- CPRI MUX and eCPRI MUX can be configured simultaneously only when both the converging and converged boards in a group are UBBPh series boards. In other cases, CPRI MUX and eCPRI MUX cannot be configured simultaneously for the converging and converged boards in a group.
- In intra-mode inter-BBU deployment scenarios, eCPRI MUX is not supported by the leaf BBU.
- For AAUs of versions earlier than SRAN16.1, the software loading control rights need to be granted to the converging party during the upgrade to SRAN16.1.
- If the converging party uses a line rate of 50 Gbit/s over the eCPRI interface, the version of the converged party must be upgraded to SRAN20.1 or later.

UMTS and LTE FDD Separate-MPT

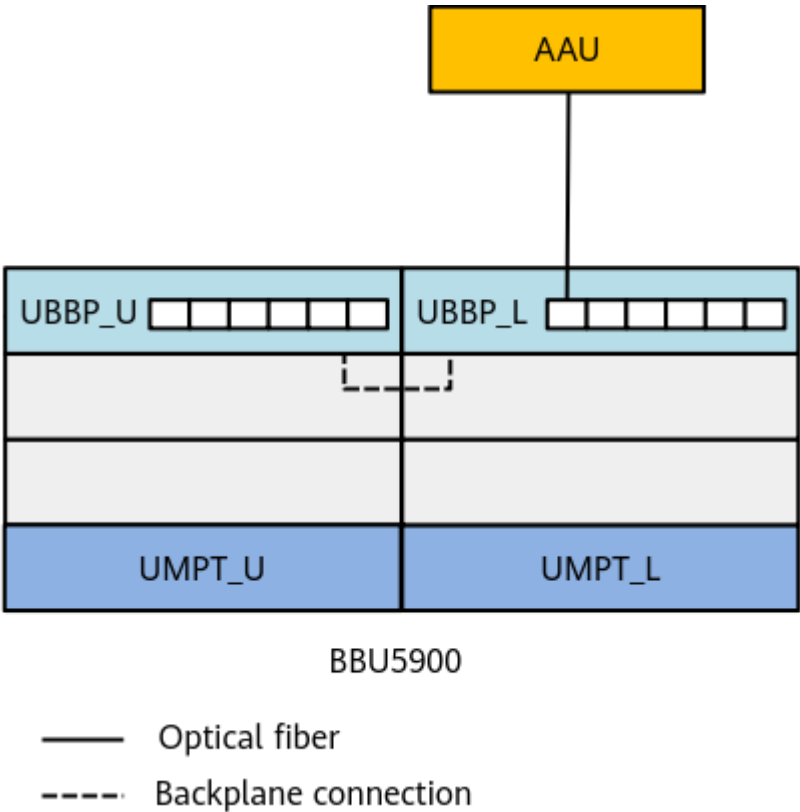
When eCPRI MUX is used in UMTS and LTE FDD separate-MPT scenarios, the AAU is connected to an LTE baseband processing unit through an optical fiber. When the **RRUCHAIN.PROTOCOL** parameter is set to **eCPRI** for a UBBPg series board serving as the converging board, and the working mode specified by the **RRU.RS** parameter is changed from excluding UMTS to including UMTS or vice versa, the board automatically resets to validate the new configuration. If the working mode specified by the **RRU.RS (LTE eNodeB, 5G gNodeB)** parameter includes UMTS, the eCPRI link of the RF module fixedly occupies the bandwidths of two UMTS cells.

The constraints on eCPRI MUX in this scenario are listed as follows:

- UMTS can only serve as the converged party, but cannot serve as the converging party.
- The converging party supports only FDD but does not support TDD.
- The converging boards must be UBBPg or UBBPh series boards, and the converged boards must be UBBP series boards.
- CPRI MUX and eCPRI MUX cannot be supported simultaneously by a converging board.
- One converged board cannot correspond to two converging boards, and two converged boards cannot correspond to one converging board.
- Only single-fiber eCPRI MUX is supported. The load sharing topology is not supported.
- A UBBPg series board serving as the converging board can forward data of a maximum of six UMTS cells. A UBBPh series board serving as the converging board can forward data of a maximum of 18 UMTS cells. (The actual number of cells whose data can be forwarded depends on the number of board ports compatible with DSFP optical modules.)

Figure 3-8 shows eCPRI MUX in a UL separate-MPT scenario.

Figure 3-8 eCPRI MUX in a UL separate-MPT scenario (using the BBU5900 as an example)



3.1.4.3 Hybrid-MPT Scenarios

In hybrid-MPT scenarios, there are both co-MPT and separate-MPT in a BBU.

In this document, hybrid-MPT indicates that UMTS, LTE FDD, and NR FDD modes are deployed in the same BBU. The BBU is configured with two main control boards, one working in UMTS mode and the other working in LTE FDD and NR FDD modes at the same time. In this case, LTE FDD and NR FDD work in co-MPT mode. UMTS works in separate-MPT mode with LTE FDD and NR FDD.

When eCPRI MUX is used in hybrid-MPT scenarios, the AAU can be connected to an LTE FDD baseband processing unit or an NR FDD baseband processing unit through an optical fiber. When the **RRUCHAIN.PROTOCOL (LTE eNodeB, 5G gNodeB)** parameter is set to **eCPRI** for a UBBPg series board serving as the converging board, and the working mode specified by the **RRU.RS (LTE eNodeB, 5G gNodeB)** parameter is changed from excluding UMTS to including UMTS or vice versa, the board automatically resets to validate the new configuration. If the working mode specified by the **RRU.RS (LTE eNodeB, 5G gNodeB)** parameter includes UMTS, the eCPRI link of the RF module fixedly occupies the bandwidths of two UMTS cells.

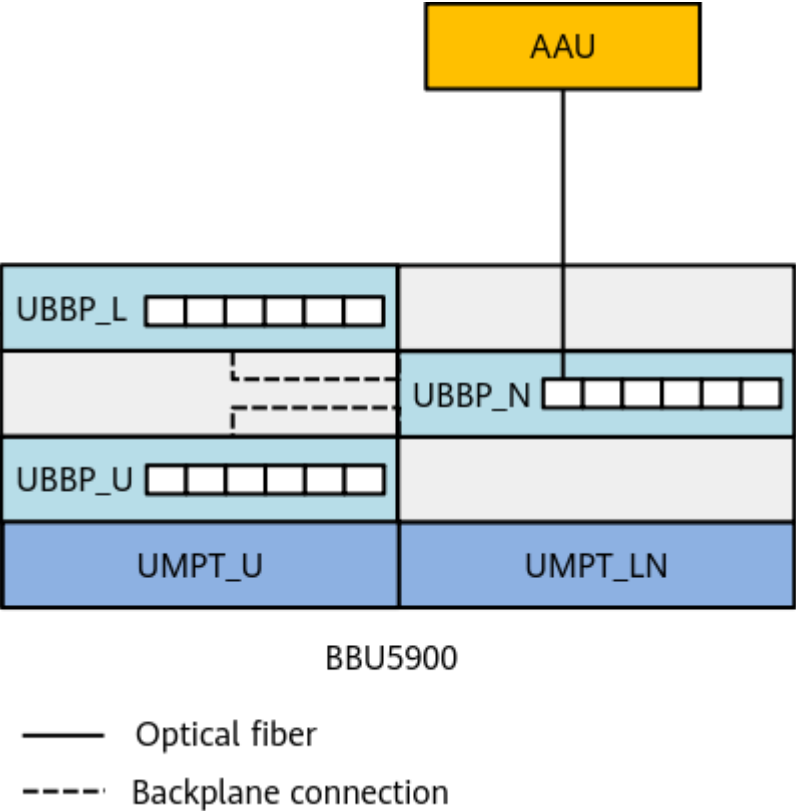
The constraints on eCPRI MUX in this scenario are listed as follows:

- UMTS can only serve as the converged party, but cannot serve as the converging party.
- The converging party supports only FDD but does not support TDD.

- The converging boards must be UBBPg or UBBPh series boards, and the converged boards must be UBBP series boards.
- CPRI MUX and eCPRI MUX cannot be supported simultaneously by a converging board.
- One converged board cannot correspond to two converging boards, and two converged boards cannot correspond to one converging board.
- Only single-fiber eCPRI MUX is supported. The load sharing topology is not supported.
- A UBBPg series board serving as the converging board can forward data of a maximum of six UMTS cells. A UBBPh series board serving as the converging board can forward data of a maximum of 18 UMTS cells. (The actual number of cells whose data can be forwarded depends on the number of board ports compatible with DSFP optical modules.)

Figure 3-9 shows eCPRI MUX in a hybrid-MPT scenario.

Figure 3-9 eCPRI MUX in a hybrid-MPT scenario (using the BBU5900 as an example)



3.1.4.4 Bandwidth Specifications of LTE and NR Fiber Sharing Cells

In eCPRI MUX scenarios, the maximum bandwidth specifications of LTE and NR fiber sharing cells are dependent on the number of antennas, numbers of uplink and downlink layers, and line rate, as listed in Table 3-15.

Table 3-15 Maximum bandwidth specifications of LTE and NR fiber sharing cells

Number of Antennas	Number of Uplink and Downlink Layers	25 Gbit/s	2x25 Gbit/s
64T64R	16 in the downlink 8 in the uplink	NR X MHz+ LTE Y MHz ^a	NR 100 MHz+LTE 5x20 MHz
32T32R	16 in the downlink 8 in the uplink	NR X MHz+ LTE Y MHz ^a	NR 100 MHz+LTE 5x20 MHz
a: $X + Y \leq 100$			

3.1.4.5 Backplane Switching Specifications

The backplane switching specifications between the converging board and converged boards in eCPRI MUX scenarios are as follows. The backplane switching specifications listed in the following table are given based on the line rate over an eCPRI interface in typical cell configurations described in [3.1.2 Line Rate Requirements in Typical Cell Configurations](#).

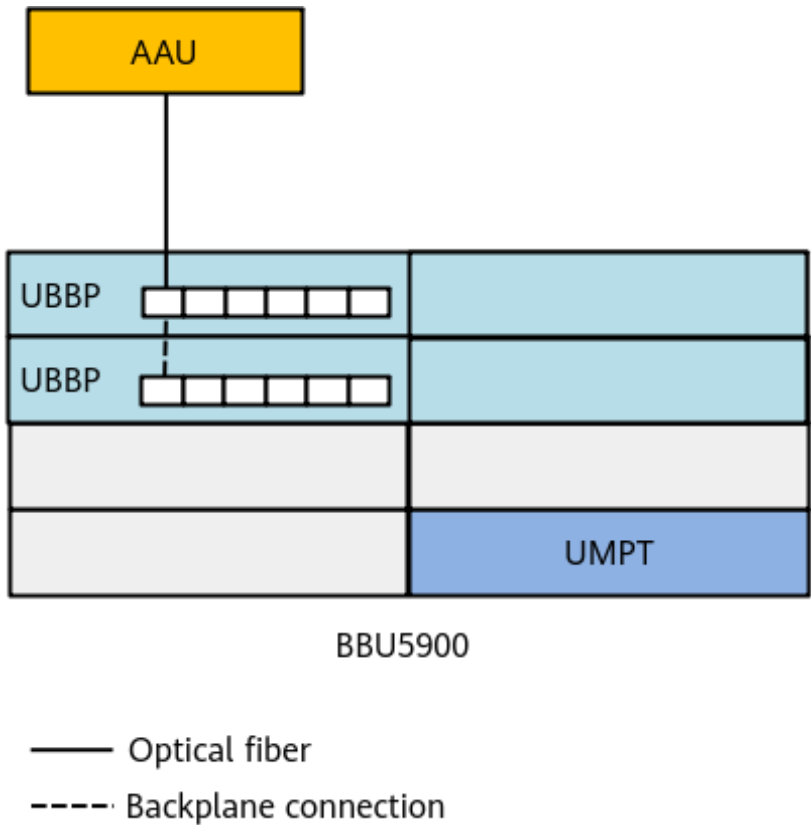
Converging Board	Converged Board	Backplane Switching Specifications
UBBPfw1	UBBPfw1	X MHz for NR + Y MHz for LTE ^a
UBBPfw1	UBBPf3/UBBPg2a/ UBBPg3/UBBPg3b/ UBBPh3/UBBPh3a/ UBBPh3b	X MHz for NR + Y MHz for LTE ^b
UBBPf3/UBBPg2a/ UBBPg3/UBBPg3b	UBBPf3/UBBPg2a/ UBBPg3/UBBPg3b/ UBBPh3/UBBPh3a/ UBBPh3b	NR X MHz+ LTE Y MHz ^b
UBBPf3/UBBPg2a/ UBBPg3/UBBPg3b/ UBBPh3/UBBPh3a/ UBBPh3b	UBBPfw1	X MHz for NR + Y MHz for LTE ^b
UBBPh3/UBBPh3a/ UBBPh3b	UBBPf3/UBBPg2a/ UBBPg3/UBBPg3b/ UBBPfw1	X MHz for NR + Y MHz for LTE ^b

Converging Board	Converged Board	Backplane Switching Specifications
UBBPh3/UBBPh3a/ UBBPh3b	UBBPh3/UBBPh3a/ UBBPh3b	X MHz for NR + Y MHz for LTE ^a
a: $X + Y \leq 400$, b: $X + Y \leq 200$		

3.1.5 eCPRI Sharing

For a single-mode base station, the eCPRI sharing mode can be used to save optical fibers between the BBU and AAU(s). The baseband processing unit that processes services does not directly connect to the AAU through an optical fiber. Instead, it shares the optical fiber connecting to the AAU with another baseband processing unit that provides the interface. The two baseband processing units are interconnected through the backplane of the BBU subrack, [Figure 3-10](#) takes the use of one baseband processing unit that processes services as an example. Actually, one baseband processing unit that provides the interface can be interconnected with multiple baseband processing units that process services.

Figure 3-10 eCPRI sharing (using the BBU5900 as an example)



The backplane switching specifications between the baseband processing unit that provides the interface and baseband processing units that process services are listed in the following table. The backplane switching specifications listed in the following table are given based on the line rate over an eCPRI interface in typical

cell configurations described in [3.1.2 Line Rate Requirements in Typical Cell Configurations](#).

Converging Board	Converged Board	Backplane Switching Specifications
UBBPfw1	UBBPfw1	The supported maximum total cell bandwidth is 400 MHz. For example, there can be four 100 MHz NR cells, eight 50 MHz NR cells, twenty 20 MHz LTE cells, or forty 10 MHz LTE cells.
UBBPfw1	UBBPf3/UBBPg2a/ UBBPg3/UBBPg3b/ UBBPh3/UBBPh3a/ UBBPh3b	The supported maximum total cell bandwidth is 200 MHz. For example, there can be two 100 MHz NR cells, four 50 MHz NR cells, ten 20 MHz LTE cells, or twenty 10 MHz LTE cells.
UBBPf3/UBBPg2a/ UBBPg3/UBBPg3b	UBBPf3/UBBPg2a/ UBBPg3/UBBPg3b/ UBBPh3/UBBPh3a/ UBBPh3b	The supported maximum total cell bandwidth is 200 MHz. For example, there can be two 100 MHz NR cells, four 50 MHz NR cells, ten 20 MHz LTE cells, or twenty 10 MHz LTE cells.
UBBPf3/UBBPg2a/ UBBPg3/UBBPg3b/ UBBPh3/UBBPh3a/ UBBPh3b	UBBPfw1	The supported maximum total cell bandwidth is 200 MHz. For example, there can be two 100 MHz NR cells, four 50 MHz NR cells, ten 20 MHz LTE cells, or twenty 10 MHz LTE cells.
UBBPh3/UBBPh3a/ UBBPh3b	UBBPf3/UBBPg2a/ UBBPg3/UBBPg3b/ UBBPfw1	The supported maximum total cell bandwidth is 200 MHz. For example, there can be two 100 MHz NR cells, four 50 MHz NR cells, ten 20 MHz LTE cells, or twenty 10 MHz LTE cells.

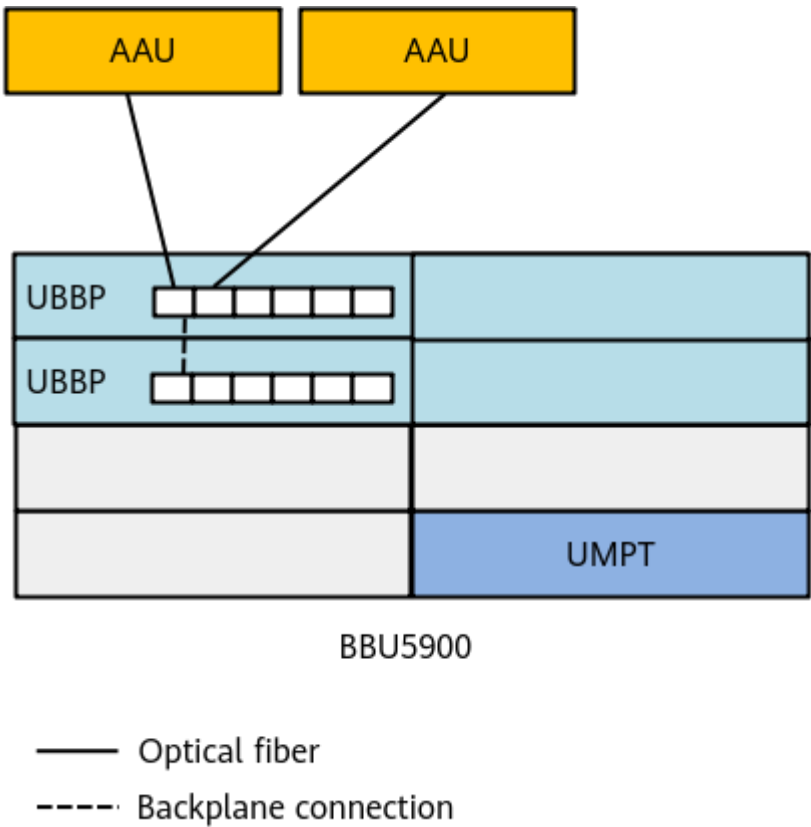
Converging Board	Converged Board	Backplane Switching Specifications
UBBPh3/UBBPh3a/ UBBPh3b	UBBPh3/UBBPh3a/ UBBPh3b	The supported maximum total cell bandwidth is 400 MHz. For example, there can be four 100 MHz NR cells, eight 50 MHz NR cells, twenty 20 MHz LTE cells, or forty 10 MHz LTE cells.

 NOTE

- It is recommended that the baseband processing unit that processes services provides optical fiber connections. Before using the eCPRI sharing mode if required, familiarize yourself with the network impacts described in [3.2.2 Impacts](#).
- In eCPRI sharing mode, when the baseband processing unit that provides the interface is faulty, the cells on the baseband processing unit that processes services will be out of service and related alarms are generated.
- eCPRI sharing and other coordination-based features share the backplane bandwidth.

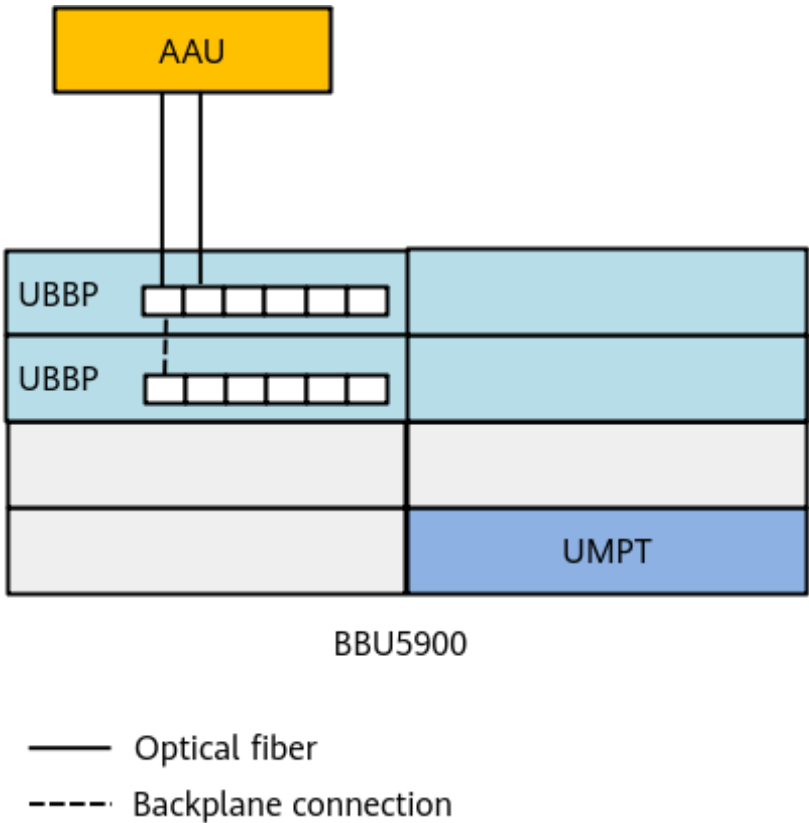
In eCPRI sharing mode, the baseband processing unit that provides the interface can be connected to the AAU(s) in a star or load sharing topology, as shown in [Figure 3-11](#) and [Figure 3-12](#).

Figure 3-11 eCPRI sharing in a star topology (using the BBU5900 as an example)



Intra-board load sharing is recommended if a load sharing topology is used. In an inter-board load sharing topology, cells must be deployed on the baseband processing units directly connected to the AAU(s) through optical fibers.

Figure 3-12 eCPRI sharing in an intra-board load sharing topology (using the BBU5900 as an example)



3.1.6 Security

The eCPRI interface transmits data over the Ethernet, which increases the risk of information leakage, tampering, and flood attacks. The following security measures are supported in base stations to ensure eCPRI interface security:

- TLS connection for authentication, encryption, and integrity protection
TLS connection is controlled by the **SSL.RRUTLSCONNTYPE** parameter. For details about the parameter values, see [Table 3-16](#). This function is used for the control plane between the BBU and AAU.
TLS authentication uses digital certificates. Asymmetric keys are used to authenticate equipment identities during digital certificate authentication. Specifically, the sender uses a private key to sign data, and the receiver uses a public key in the certificate to verify signature validity. Both the receiver and the sender confirm each other's identity by checking digital certificates to ensure the integrity of transmitted data and identify the confidentiality of end-to-end interconnection. For details, see *SSL* and *PKI*.

Table 3-16 TLS connection type parameter value

Parameter Value	Meaning	Description
TLS(Only TLS)	Only RRU's using the TLS protocol can be connected to the BBU.	This value is the default value.
PLAINTEXT(Only PlainText)	Only RRU's using plaintext can be connected to the BBU.	All AAUs connected to the BBU must support plaintext-based access. Otherwise, the negotiation fails. Unauthorized clients may access NEs.
TLS_PLAINTEXT(TLS and PlainText)	RRU's using plaintext or the TLS protocol can be connected to the BBU.	Unauthorized clients may access NEs.

If TLS connection negotiation fails on the eCPRI interface, ALM-26235 RF Unit Maintenance Link Failure is reported.

NOTE

- For baseband processing units and RF modules with eCPRI ports, the Huawei initial certificate preconfigured before delivery is for site deployment only. After the site deployment, the operator-issued device certificate and the operator's root certificate need to be obtained from the operator's public key infrastructure (PKI) system. For details, see *PKI*.
- If the Huawei initial certificate of an AAU is damaged, you can run the **SET RRUSECPOLICY** command on the BBU to change the authentication mode to **NONE(Verify None)**. You can also run the **DSP RRUTLSERRINFO** command to query the actual negotiated authentication mode.
- In UMTS and LTE FDD separate-MPT and hybrid-MPT scenarios, the converged party uses the certificate of the converging party.
- If the **SET RRUSECPOLICY** command is executed on the BBU side with the **SSL.RRUTLSCONNTYPE** parameter set to **PLAINTEXT(Only PlainText)** or **TLS_PLAINTEXT(TLS and PlainText)**, AAUs using plaintext can be connected to the BBU. If the negotiation fails because AAUs are not allowed to use plaintext, the BBU reports ALM-26219 RF Unit Maintenance Link Negotiation Failure.
- MACsec connection for authentication, encryption, and integrity protection
This function is disabled by default. It protects control-, user-, and synchronization-plane data between the BBU and AAU. MACsec connection is controlled by the **CPRIPORT.MACSECCONNTYPE (LTE eNodeB, 5G gNodeB)** parameter. For details, see *MACsec*.
- PDCCP encryption over the air interface
This function is enabled by default. It prevents the user-plane data between the BBU and AAU from being illegally intercepted or disclosed. For details, see *Radio Security*.
- Anti-flooding function
This function is enabled by default. Currently, it is supported only on BBU eCPRI interfaces. The detection of common protocols (such as TCP/UDP/ARP)

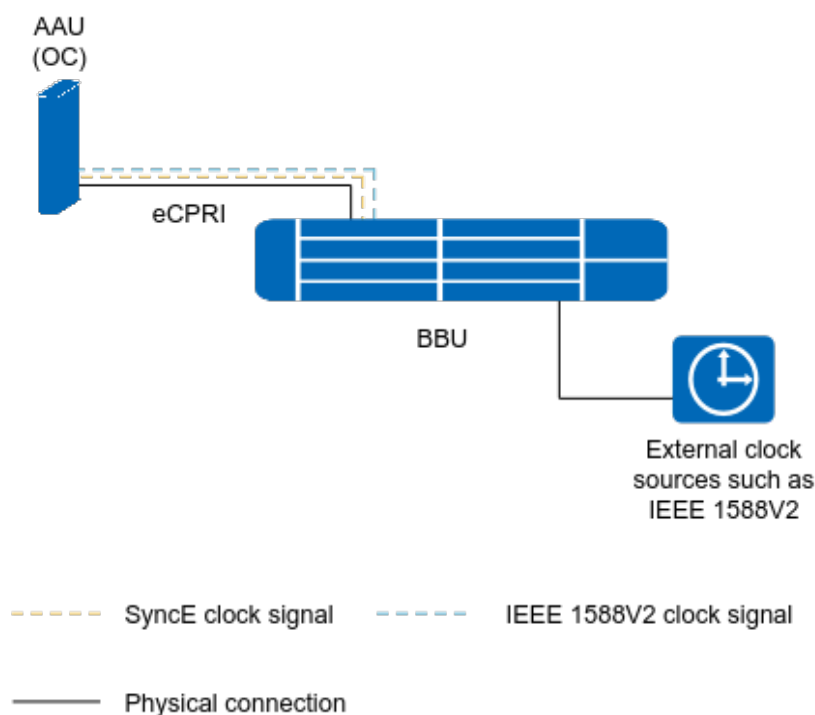
is supported. If the number of detected packets exceeds a specified threshold, ALM-25950 Base Station Being Attacked is reported.

3.1.7 Clock Synchronization

When the BBU and AAU are connected through the eCPRI port, a base station provides IEEE 1588V2+SyncE clock signals for the AAU. SyncE refers to synchronous Ethernet. The IEEE 1588V2 clock and SyncE clock share the same source, and the two types of clocks implement time synchronization and frequency synchronization, respectively. The introduction of the SyncE clock can improve the frequency synchronization precision of the AAU to meet precision requirements.

The BBU is configured as the IEEE 1588V2+SyncE master, and the AAU is used as the IEEE 1588V2+SyncE slave to implement synchronization with the BBU. The configuration does not require any extra license. The AAU can work in ordinary clock (OC) mode only, as shown in [Figure 3-13](#).

Figure 3-13 Example of IEEE 1588V2 and SyncE combination



Clock Requirements

The AAU obtains synchronous clock signals from the BBU through the eCPRI port. In addition, it obtains only the 1588V2+SyncE clock signals.

The requirements for eCPRI fronthaul clock synchronization specifications are as follows:

- For time synchronization accuracy requirements, see section 4.2 "Timing accuracy requirements" in eCPRI Transport Network V1.2.

- For frequency synchronization accuracy requirements, see section 6.4.2 "Frequency synchronization requirements" in 802.1CM-2018 - IEEE Standard for Local and metropolitan area networks -- Time-Sensitive Networking for Fronthaul.

The BBU functions as the IEEE 1588V2+SyncE master, and the delivered IEEE 1588V2 message contains the clock quality level. The AAU functions as the IEEE 1588V2+SyncE slave. According to IEEE 1588V2, the clock link is selected and switched over based on the clock link status and the clock quality level in the message. The switchover interrupts services for a short period of time.

 NOTE

In CPRI mode, RF modules can switch over the clock link only based on the clock link status.

Impact of the Clock Mutual Lock Switchover

The impact of BBU clock mutual lock switchover in the eCPRI scenario is the same as that in the CPRI scenario. For details, see "Inter-RAT Clock Lock" in *Common Clock*.

3.2 Network Analysis

3.2.1 Benefits

This feature enables the reduction of the transmission bandwidth between the BBU and AAU as well as the transmission resources required by massive MIMO and other large-bandwidth multi-antenna services. This further saves costs.

3.2.2 Impacts

The impacts between this function and other functions are described in [3.3.2.3 Function Impacts](#).

- eCPRI MUX in separate-MPT scenarios
 - Operations or exceptions of the converging party

Data and voice services of the converged party will be interrupted if any of the following occurs on the converging party. The interruptions last less than 3 minutes in most cases and will not exceed 3.5 minutes.

 - A software reset is performed on the converging board or the main control board, the converging board or the main control board is reset through a power cycle, or the converging board or the main control board is removed and then reinserted.
 - The converging board is blocked.
 - Commissioning is performed on a CPRI port of the converging board.
 - A CPRI port of the converging board becomes faulty.
 - Operations or exceptions of the converged party

- If a software reset is performed on the main control board of the converged party, the rate of data services of the converging party may decrease. The impact lasts for less than 3s in most cases and does not last for over 10s. However, voice services will not be affected.
- If the main control board of the converged party is reset through a power cycle or removed and then inserted, data and voice services of the converging party will be interrupted for less than 1 minute in most cases. In the worst case, the service interruption does not last for over 3 minutes.
- eCPRI MUX in co-MPT scenarios
Data and voice services of the converged party will be interrupted if any of the following occurs on the converging party. Most interruptions last less than 3 minutes and the longest does not exceed 3.5 minutes.
 - A software reset is performed on the converging board, the converging board is reset through a power cycle, the converging board is blocked, or the converging board is removed and then reinserted.
 - Commissioning is performed on a CPRI port of the converging board.
 - A CPRI port of the converging board becomes faulty.
 - After enhanced eCPRI-based inter-board load sharing takes effect, the peak rate of services carried on the other optical fiber may be affected.
- eCPRI sharing scenarios
Data and voice services of the converged party will be interrupted if any of the following occurs on the baseband processing unit that provides the interface. The interruptions last less than 3 minutes in most cases and will not exceed 3.5 minutes.
 - A software reset is performed on the converging board, the converging board is reset through a power cycle, the converging board is blocked, or the converging board is removed and then reinserted.
 - Commissioning is performed on a CPRI port of the converging board.
 - A CPRI port of the converging board becomes faulty.

 NOTE

Service interruption duration does not include the duration of manual operations, such as blocking a board or removing and then inserting a board.

3.3 Requirements

3.3.1 Licenses

None

3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
NR FDD Low-frequency NR TDD High-frequency NR TDD	Network Synchronization	None	<i>Synchronization</i>
LTE TDD LTE FDD	Enhanced Synchronization	None	<i>Synchronization in eRAN Feature Documentation</i>

3.3.2.2 Mutually Exclusive Functions

None

3.3.2.3 Function Impacts

None

3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

The main control board must be a UMPTe, UMPTg, UMPTga, UMPTHa, or UMPTHb. However, in intra-RAT inter-BBU scenarios, the eCPRI module can be deployed if the main control board in the root BBU is a UMPTe, UMPTg, UMPTga, UMPTHa, or UMPTHb.

BBPs support the eCPRI protocol type. For details about the protocol type of a BBP, see descriptions about port types in the BBU hardware description under **Hardware Description of Description** in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Only AAUs support this function. For details about AAU models that support the eCPRI protocol, see CPRI port specifications in the AAU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

3.3.4 Others

None

3.4 Operation and Maintenance

3.4.1 Data Configuration

3.4.1.1 Data Preparation

By default, the security and clock synchronization functions for the eCPRI interface are enabled and do not need to be manually configured.

For eCPRI topologies (including eCPRI MUX and eCPRI sharing), the parameters in the **RRUCHAIN** and **RRU** MOs need to be set for the eCPRI interface. For detailed configurations, see section "Operation and Maintenance" for a specific base station model in *RF Unit and Topology Management*. For co-MPT scenarios:

Parameter Name	Parameter ID	Setting Notes
eCPRI Inter-Board Enhanced Switch	EQUIPMENT.ECPRIIBEN HSW (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.

For eCPRI MUX in separate-MPT and hybrid-MPT scenarios:

- The AAU and corresponding chain/ring configurations of the converged party take effect only after those of the converging party have been configured.
- There are no special requirements on the configurations of the converging party. For detailed configurations, see section "Operation and Maintenance" for a specific base station model in *RF Unit and Topology Management*.
- For the converged party, most of the configurations are the same as those described in section "Operation and Maintenance" for a specific base station model in *RF Unit and Topology Management*. The following table lists the parameter configurations that are different.

Parameter Name	Parameter ID	Setting Notes
Access Type	RRUCHAIN.A7	Set this parameter to PEERPORT .

Parameter Name	Parameter ID	Setting Notes
Head Cabinet No.	RRUCHAIN.HCN	Set this parameter to the cabinet number of the baseband processing unit that provides the peer CPRI port.
Head Subrack No.	RRUCHAIN.HSRN	Set this parameter to the subrack number of the baseband processing unit that provides the peer CPRI port.
Head Slot No.	RRUCHAIN.HSN	Set this parameter to the slot number of the baseband processing unit that provides the peer CPRI port.
Head Port No.	RRUCHAIN.HPN	Set this parameter to the optical port number of the peer CPRI port.
Local Slot No.	RRUCHAIN.LSN	Set this parameter to the slot number of the converged board.

The following table describes the parameters related to cell activation.

Parameter Name	Parameter ID	Setting Notes
CPRI Compression	Cell.CPRICompression	Retain the default value NO_COMPRESSION for an LTE FDD or NB-IoT cell as eCPRI compression is not required. Set this parameter to ECPRI_COMPRESSION for an LTE TDD cell if eCPRI compression is required.
CPRI Compression	NRDUCellTrp.CpriCompression	Set this parameter to 2_COMPRESSION for an NR FDD cell. Set this parameter to 3DOT2_COMPRESSION for an NR TDD cell.

Parameter Name	Parameter ID	Setting Notes
Alarm Switch	gNBOamParam.AlarmSwitch	It is recommended that the ECPRI_CPRI_INSUFT_ALM_RPT_SW option be selected in NR 8T8R and massive MIMO scenarios.

3.4.1.2 Using MML Commands

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Activating eCPRI compression for an LTE TDD massive MIMO cell
MOD CELL: LocalCellId=0, CPRICompression=ECPRI_COMPRESSION;
//Activating eCPRI compression for other TDD cells served by the AAU serving the cell
MOD CELL: LocalCellId=1, CPRICompression=ECPRI_COMPRESSION;
//Enabling enhanced eCPRI-based inter-board load sharing
SET EQUIPMENT: ECPRIIBENHSW=ON;
//Configuring parameters for activating an NR FDD cell
MOD NRDUCELLTRP: NrDuCellTrpId=0, TrpType=DEFAULT, CpriCompression=2_COMPRESSION;
//Configuring parameters for activating an NR TDD cell
MOD NRDUCELLTRP: NrDuCellTrpId=0, TrpType=DEFAULT, CpriCompression=3DOT2_COMPRESSION;
//(Optional) Selecting the ECPRI_CPRI_INSUFT_ALM_RPT_SW option in NR 8T8R and massive MIMO scenarios
MOD GNBOAMPARAM: AlarmSwitch=ECPRI_CPRI_INSUFT_ALM_RPT_SW-1;
```

Deactivation Command Examples

```
//Disabling enhanced eCPRI-based inter-board load sharing
SET EQUIPMENT: ECPRIIBENHSW=OFF;
//Deactivating eCPRI compression
MOD CELL: LocalCellId=0, CPRICompression=NO_COMPRESSION;
MOD CELL: LocalCellId=1, CPRICompression=NO_COMPRESSION;
//Configuring parameters for deactivating an NR FDD cell
MOD NRDUCELLTRP: NrDuCellTrpId=0, TrpType=DEFAULT, CpriCompression=NO_COMPRESSION;
//Configuring parameters for deactivating an NR TDD cell
MOD NRDUCELLTRP: NrDuCellTrpId=0, TrpType=DEFAULT, CpriCompression=NO_COMPRESSION;
//(Optional) Deselecting the ECPRI_CPRI_INSUFT_ALM_RPT_SW option
MOD GNBOAMPARAM: AlarmSwitch=ECPRI_CPRI_INSUFT_ALM_RPT_SW-0;
```

3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

3.4.2 Activation Verification

eCPRI Interface Security

Run the **DSP RRUTLSERRINFO** command to query TLS information. If the value of **TLS Connection State** is **Normal**, TLS authentication has been enabled.

```
+++ 0 2022-12-26 15:08:50
O&M #123
%%DSP RRUTLSERRINFO: CN=0, SRN=100,%%
RETCODE = 0 Operation succeeded.

Display RRU TLS Information
-----
Cabinet No. = 0
Subrack No. = 100
Slot No. = 0
Connection Type = TLS
TLS Connection State = Normal
TLS Version = TLS1.3
TLS Protocol = Diffie-Hellman Group: SECP256R1; Signature Algorithm: RSA_PSS_RSAE_SHA256; Cipher Suite: TLS_AES_256_GCM_SHA384;
TLS Authentication Mode = Verify Peer Certificate
Cert Name = appcert.pem
Error Code = NULL
Error Description = NULL
Certificate Verify Result = NULL
Certificate Verify Result Description = NULL
Occurrence Time = 2022-12-08 15:32:09
(Number of results = 1)

--- END
```

Run the **DSP CPRIPORT** command to query MACsec-related information. If the value of **MACsec Connection Type** is **MACSEC** and the value of **MACsec Connection State** is **NORMAL**, MACsec authentication has been enabled.

```
To query the dynamic information about a CPRI port, run the following command:
DSP CPRIPORT:
The result is shown as follows:
+++ 0 2022-12-17 08:00:00
O&M #41
%%DSP CPRIPORT:%%
RETCODE = 0 Operation succeeded.

Display CPRIPort Dynamic Information
-----
Cabinet No. Subrack No. Slot No. Port Type Port No. Sub Port No. Install Status OFF Module Type Administrative Status CPRI Bit Rate(kbit/s) CPRI Port Mode CPRI Port Line Status Availability Status Remaining Continuous Bandwidth(kbit/s) Remaining Discontinuous Bandwidth(kbit/s) Protocol Type Cable Reservation Status Network Cable Length (m)
0 0 3 CPRI 0 0 Not In Position NULL Unlocked NULL NULL Normal Normal NULL NULL CPRI NULL NULL
0 0 3 CPRI 1 0 Not In Position NULL Unlocked NULL NULL Normal Normal NULL NULL CPRI NULL NULL
0 0 3 CPRI 2 0 In Position DIFF DIFF Unlocked 2.5G Master Normal Normal 1493.10 0 CPRI NULL NULL
0 0 3 CPRI 3 0 Not In Position NULL Unlocked NULL NULL Normal Normal NULL NULL CPRI NULL NULL
0 0 3 CPRI 4 0 Not In Position NULL Unlocked NULL NULL Normal Normal NULL NULL CPRI NULL NULL
0 0 3 CPRI 5 0 Not In Position NULL Unlocked NULL NULL Normal Normal NULL NULL CPRI NULL NULL
(Number of results = 6)

Display CPRIPort MACsec Information
-----
Cabinet No. = 0
Subrack No. = 0
Slot No. = 3
Port No. = 2
Sub Port No. = 0
MACsec Connection Type = MACSEC
MACsec Connection State = NORMAL
MAC-TLS Version = ESP-TLS-0
MACsec Cipher Suite = Diffie-Hellman Group: SECP256R1;
MACsec Certificate Name = appcert.pem
MACsec Error Code = 0
MACsec Error Description = NULL
MACsec Certificate Verify Result = 0
MACsec Certificate Verify Result Description = NULL
MACsec Connection Occurrence Time = 2022-12-17 08:00:00
(Number of results = 1)

--- END
```

eCPRI Clock Synchronization

Run the **DSP CLKSTAT** command to check the system clock status. If **Current Clock Source** is **SyncEth Clock+IP Clock**, **Current Clock Source State** is **Normal**, and **PLL Status** is **Locked** in the command output, the system clock has been configured.

```
+++ Ecpril45      2018-11-02 15:02:22
O&M #27936
%%/*1879050245*/DSP CLKSTAT:SRN=60;%%
RETCODE = 0 Operation succeeded.

Display System Clock Status
-----
                Cabinet No. = 0
                Subrack No. = 60
                Slot No. = 0
        Current Clock Source = SyncEth Clock+IP Clock
Current Clock Source State = Normal
                PLL Status = Locked
(Number of results = 1)

--- END
```

eCPRI Topologies (Including eCPRI MUX and eCPRI Sharing)

Run the **DSP BRD** command to check whether the AAU status is normal. If **Availability Status** is **Normal**, the AAU is available and the eCPRI topology has been configured.

```
+++ Ecpril45      2018-11-02 15:04:56
O&M #27958
%%/*1879050247*/DSP BRD:;%%
RETCODE = 0 Operation succeeded.

Display Board
-----
Cabinet No. Subrack No. Slot No. Config Type SubBoard Type Administrative State Standby Status Operational State Alarm Status Availability Status WorkMode
0 0 0 0 VBBP-W NULL Unblocked NULL Enabled Major Normal NULL
0 0 0 7 UMPT NULL NULL Active Enabled Normal Normal NORMAL
0 0 0 16 FAN NULL NULL NULL Enabled Normal Normal NULL
0 0 0 18 UFEU NULL NULL NULL Disabled Major Uninstalled NULL
0 0 0 19 UFEU NULL NULL NULL Enabled Normal Normal NULL
0 0 60 0 AIRU NULL Unblocked NULL Enabled Normal Normal NULL
(Number of results = 6)

--- END
```

LTE TDD eCPRI Compression

Run the **LST CELL** command. If the value of **CPRI Compression** is **ECPRI_COMPRESSION** in the command output, the eCPRI compression function has been activated.

3.4.3 Network Monitoring

eCPRI Port Performance Counters

The performance of an eCPRI port can be monitored by checking the following counters.

Counter Name	Counter Description
VS.ECPRIPORT.TxBytes (5G gNodeB, LTE eNodeB)	Number of Bytes Transmitted from an eCPRI Port
VS.ECPRIPORT.RxBytes (5G gNodeB, LTE eNodeB)	Number of Bytes Received by an eCPRI Port
VS.ECPRIPORT.TxPackets (5G gNodeB, LTE eNodeB)	Number of Packets Transmitted from an eCPRI Port
VS.ECPRIPORT.RxPackets (5G gNodeB, LTE eNodeB)	Number of Packets Received by an eCPRI Port
VS.ECPRIPORT.TxMaxSpeed (5G gNodeB, LTE eNodeB)	Maximum Transmit Rate of an eCPRI Port
VS.ECPRIPORT.TxMinSpeed (5G gNodeB, LTE eNodeB)	Minimum Transmit Rate of an eCPRI Port
VS.ECPRIPORT.TxMeanSpeed (5G gNodeB, LTE eNodeB)	Average Transmit Rate of an eCPRI Port
VS.ECPRIPORT.RxMaxSpeed (5G gNodeB, LTE eNodeB)	Maximum Receive Rate of an eCPRI Port
VS.ECPRIPORT.RxMinSpeed (5G gNodeB, LTE eNodeB)	Minimum Receive Rate of an eCPRI Port
VS.ECPRIPORT.RxMeanSpeed (5G gNodeB, LTE eNodeB)	Average Receive Rate of an eCPRI Port
VS.ECPRIPORT.TxTotalBW (5G gNodeB, LTE eNodeB)	Total Transmit Bandwidth of an eCPRI Port
VS.ECPRIPORT.RxTotalBW (5G gNodeB, LTE eNodeB)	Total Receive Bandwidth of an eCPRI Port
VS.ECPRIPORT.RxErrPackets (5G gNodeB, LTE eNodeB)	Number of Error Packets Received by an eCPRI Port

eCPRI BER Monitoring

eCPRI bit error rate (BER) monitoring is supported on the MAE-Access. The BER types include pre-FEC BER, post-FEC BER, and PCS BER. The operations are as follows:

- Step 1** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.
 - Step 2** On the **Signaling Trace Management** tab page, choose **Base Station Device and Transport > Device Monitoring > eCPRI BER Monitoring**.
 - Step 3** Create an eCPRI BER monitoring task.
 - Step 4** View the BER monitoring results.
- End

4 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

5 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- eCPRI Specification: "Common Public Radio Interface: eCPRI Interface Specification"
- eCPRI Transport Network: "Common Public Radio Interface: Requirements for the eCPRI Transport Network"
- 802.1CM-2018 - IEEE Standard for Local and metropolitan area networks -- Time-Sensitive Networking for Fronthaul
- *RF Unit and Topology Management*
- *Common Clock*
- *SSL*
- *PKI*
- *Synchronization and Radio Security in 5G RAN Feature Documentation*
- *Synchronization and Radio Security in eRAN Feature Documentation*
- *MACsec*
- *AAU Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*